Friedrich-Alexander-Universität
Technische Fakultät

1. Prologue

Knowledge Discovery in Databases with Exercises

Dominik Probst, dominik.probst@fau.de
Computer Science 6 (Data Management), Friedrich-Alexander-Universität Erlangen-Nürnberg
Summer semester 2026

Who? - Lecturer and Tutors



Dominik Probst, M.Sc.
Lecturer and Primary Contact

- Ph.D. candidate @CS6 (Data Management)
- E-Mail: `dominik.probst@fau.de`
- Website: `https://www.cs6.tf.fau.eu/dp`

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Side Note: I am not a Dr. or Prof.!

Please address me as *„Mr. Probst“*, as I would otherwise be obliged to correct you each time in order to avoid assuming a title I do not hold.

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Tutors:
Exercise Supervision

- Shree Varshan Babu
- Mahmoud Elboghdady
- Hanna Septiani
- Mohammad Salahuddin Parvez
- Olga Zotova

For whom? - Courses of Study

- **Knowledge Discovery in Databases with Exercises (KDDmUe)** - 5 ECTS
  - B.Sc./M.Sc. Data Science
  - B.Sc./M.Sc. Computer Science
  - M.Sc. International Information Systems
  - M.Sc. Medical Engineering
  - M.Sc. Information and Communication Technology
  - Possibly other courses of study (clarify with your examination office)

Important: The „Lecture Only“ Is No Longer Offered!

Until SS2023, we offered the module KDD (2.5 ECTS - only lecture) for some degree programmes. This module is no longer offered!

For whom? - Prerequisites

- **Useful prerequisites:**
  - Successful completion of the module „Einführung in Datenbanken“ (EDB)
    - Or a similar course teaching the basics of databases and SQL
  - Experience with:
    - Python
    - Jupyter Notebooks
    - Numpy
    - Pandas
    - Algorithms
    - Data structures

What? - Goal and Topics

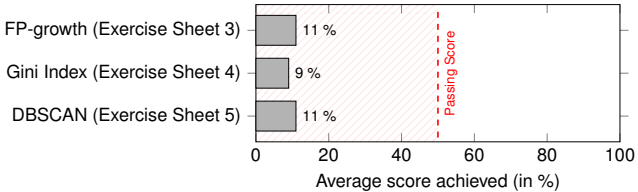
- **Goal of the module:**
  - Introduce you to the principles of data mining.
    $\Rightarrow$ This is the core of knowledge discovery in databases
- **Topics in the lecture:**
  1. Introduction
  2. Data
  3. Preprocessing
  4. Data Warehousing
  5. Mining Frequent Patterns
  6. Classification
  7. Cluster Analysis
  8. Outlier Analysis
- **Topics in the exercise:**
  1. Introduction to python and pandas (optional)
  2. Data Analysis and Data Preprocessing
  3. Frequent Patterns
  4. Classification
  5. Clustering
- **Topics in the submissions:**
  1. Frequent Patterns
  2. Classification
  3. Clustering

What? - Exercises

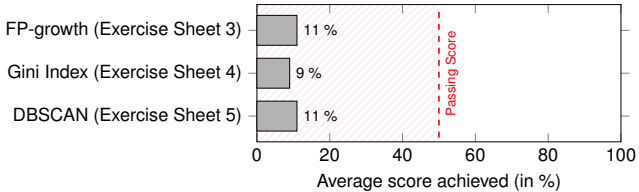
- **Exercise sessions (in presence):**
  - Working together on exercise sheets
  - Either practical data science tasks or theoretical exercises
    (varies depending on the exercise sheet)
  - Required tools:
    - Laptop capable of running Jupyter Notebooks
      (preferably one per person, one per small group is also possible)
  - Expected preparation:
    - Good understanding of the lecture content
    - Completed "Preparation" section (see exercise sheets)

- **Summer Semester 2024:**
  - Very low attendance in the exercise sessions (below 5% of registered students)
    $\Rightarrow$ Tasks based on the exercise sheets with very low point averages in the exam

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Important: The Exercises Are Crucial For The Exam!

While the lectures give you a basic theoretical overview, the exercises go into more detail, give you some more practical insights, and will be the basis of multiple questions in the exam!

What? - Submissions

- **Programming tasks (to be done at home):**
  - Implementation of individual algorithms known from the lecture for a deeper understanding
    - Programming language: Python
    - Topics: Frequent Patterns, Classification, and Clustering
  - Have to be submitted to a GitHub classroom
    - Calculation of points performed automatically after each push
    - Improvements possible at any time until the submission deadline
  - Work in small groups (up to three persons) is permitted
    - We conduct random checks for plagiarism across groups
      $\Rightarrow$ In cases where plagiarism is detected, all groups involved will receive zero points.

Participating In Or Passing The Submissions Is Not Mandatory!

However, if you get at least 50% of the points, you will receive access to a mock exam with solutions. Also some exam questions will be easier to solve if you have completed the submissions.

What? - Exam

- **Knowledge Discovery in Databases with Exercise (KDDmUe)** - Written Exam
  - Duration: 90 minutes
  - Content: Questions on lecture, exercise and submission content
  - Language: English
  - Permitted helpers: Single-sided, handwritten DIN A4 cheat sheet
  - Not permitted: Electronic devices (including calculators)

Important: Do Not Forget To Register!

Without exception[1], we can only examine participants who have also registered for this exam at the examination office.

- **(Provisional) Registration Period:** May 18 (00:01) until June 7, 2026 (23:59)
- **Stay updated at:** `https://www.cs6.tf.fau.de/kdd/reg_date`

[1] If you have to retake the exam, you also have to register manually for the exam **during** the registration period. There is no automatic registration for retakes anymore!

When? - Dates

- **Lecture** - Start in Calendar Week 16 (Today)
  - Wednesday, 08:15 - 09:45 (H20)
    (*Live stream and recording avbl. via StudOn*)
    Lecturer: Dominik Probst

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- **Guest lecture** - One time: July 10
  - Friday, 10:45 - 12:00 (H18)
    Lecturer: Daniel Krauss
    (Co-Founder and CIO @Flix)

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FLIXBUS FLIXTRAIN

- **Exercises** - Start in Calendar Week 17
  - Group 1: Monday, 16:15 - 17:45 (H4)
    Tutor: Hanna Septiani
  - Group 2: Tuesday, 12:15 - 13:45 (H4)
    Tutor: Mohammad Salahuddin Parvez
  - Group 3: Wednesday, 14:15 - 15:45 (H4)
    Tutor: Olga Zotova
  - Group 4: Thursday, 10:15 - 11:45 (H4)
    Tutor: Shree Varshan Babu
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Registration For The Exercises

Registration for exercises is mandatory to ensure an appropriate support to questions regarding setting of exercises should they arise. Registration opens at April 15th, 10:00 via StudOn.

When? - Preliminary Schedule

Calendar Week	Lecture	Exercise	Submission
16	Prologue + Introduction		
17	Data	Introduction to Python & pandas (optional)	
18	Preprocessing	Data Analysis & Data Preprocessing	
19			
20	Data Warehousing		
21	Frequent Pattern		
22		Frequent Pattern	
23	Classification		Frequent Pattern
24		Classification	
25	Cluster Analysis		Classification
26			
27	Outlier Analysis + FLiX	Clustering	Clustering
28			
29	Exam Q&A		

Where? - StudOn

- Register at:
  `https://www.studon.fau.de/studon/go/crs/6792338`
- Main source for resources. E.g.:
  - Lecture slides
  - Exercise sheets
  - Submission sheets
  - Forum
- Membership required to receive important updates on KDD
- Questions should be asked here (StudOn Forum)



Where? - GitHub

- Public repository at: `https://github.com/FAU-CS6/KDD`
- Version control of our resources including:
  - Lecture slides
  - Exercise sheets
  - Submission sheets



Help Appreciated: Error Corrections

Even though we strive for error-free lecture slides and practice sheets, there is still the possibility that errors have slipped in. You can help us mitigate these inaccuracies: Mail us, or better yet, in the case you have a GitHub account, open up a GitHub issue or create a pull request. Any pointers to errors are very much appreciated.

Any questions about this chapter?

Ask them now or ask them later in our forum:



https://www.studon.fau.de/studon/goto.php?target=lcode_kh3JHJdN

Appendix

What? - Literature

- **This lecture is based on the book by Han et al.:**
  - J. Han et al., *Data Mining: Concepts and Techniques*, 3rd. San Francisco, CA, USA: Morgan Kaufmann Publishers Inc., 2011, ISBN: 0123814790
  - Copies are available at the Science and Technology Branch Library (TNZB).
  - Lecture slides are based on slides provided by Jiawei Han with modifications by Prof. Dr.-Ing. Klaus Meyer-Wegener, Luciano Melodia, and Melanie Sigl.
  - Lecture slides have been modified and extended a lot since then.
- **Further books on this topic include, but not limited to:**
  - A. Geŕon, *Hands-on machine learning with Scikit-Learn and TensorFlow : concepts, tools, and techniques to build intelligent systems*, 2nd ed. O'Reilly Media, 2017, ISBN: 978-1491962299
  - H. Du, *Data Mining Techniques and Applications: An Introduction*. Cengage Learning EMEA, May 2010, p. 336, ISBN: 978-1844808915
  - I. H. Witten et al., *Data Mining, Fourth Edition: Practical Machine Learning Tools and Techniques*, 4th. San Francisco, CA, USA: Morgan Kaufmann Publishers Inc., 2016, ISBN: 0128042915